
AMBIENT
SYSTEMS
Climate At Your Service



Indoor Farming requires coordination of complex parameters, such as lighting, moisture, and CO2, requiring constant monitoring and adjustments to ensure HIGH-QUALITY output

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Too Many Manual Systems = Labor Intensive

CO2 Supply unit

Lighting

Form factors & Growing Systems



Environmental Control Unit

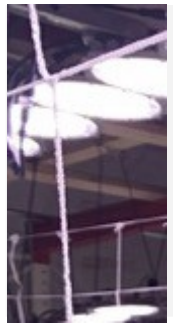
Air Conditioning

Nutrient Supply & Control Unit

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Savings Potential Through Increased Human Productivity

Utilities
$3



Rent
$30



Human Capital
$300



Only 51% of Indoor Farms Are Profitable (Agrilyst, 2019)

Undesired Interactions Between Building Conditions and Crops

(External Building Conditions)

Temperature

(Vertical Racks Ecosystem)

Soil Moisture

Lighting

Nutrients



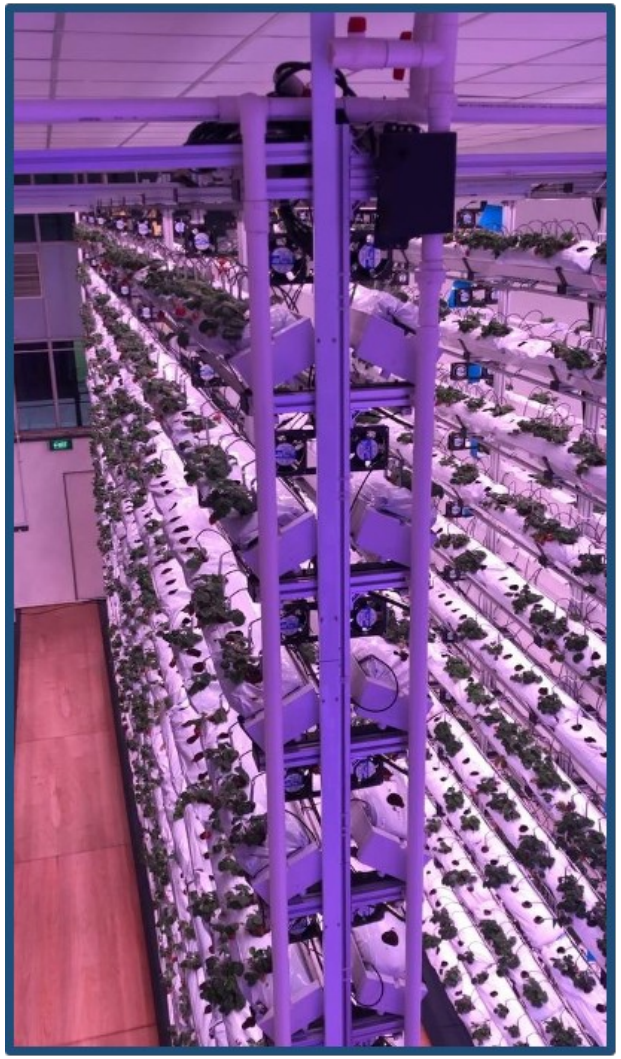
Humidity

Building Conditions affect crops

- Inconsistent Environment
- Habitat for Mold and Pest
- Affect level of photosynthesis

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Undesired Interactions Between Building Conditions and Crops



Strawberries growing in varying levels of
Temperature
Humidity
CO2 Levels



Mold Formation

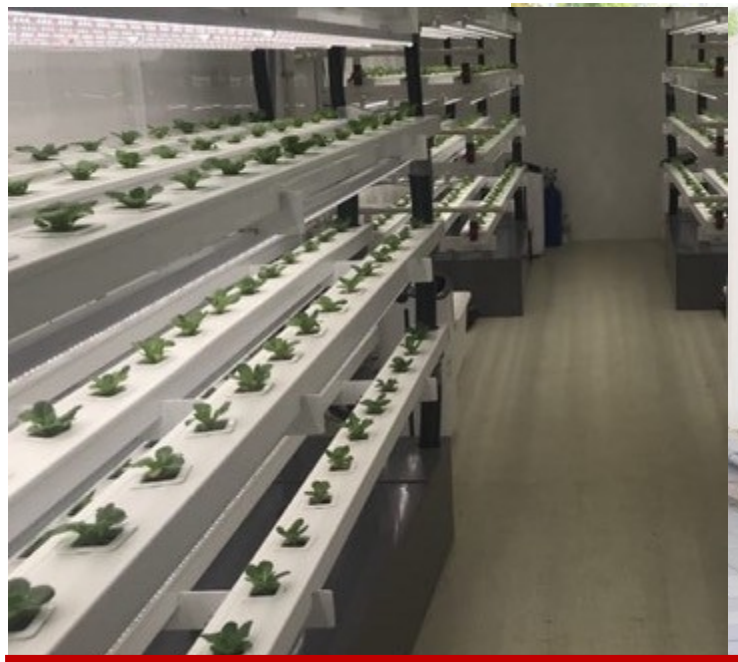


Habitat for pests

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Current Solutions are Lacking

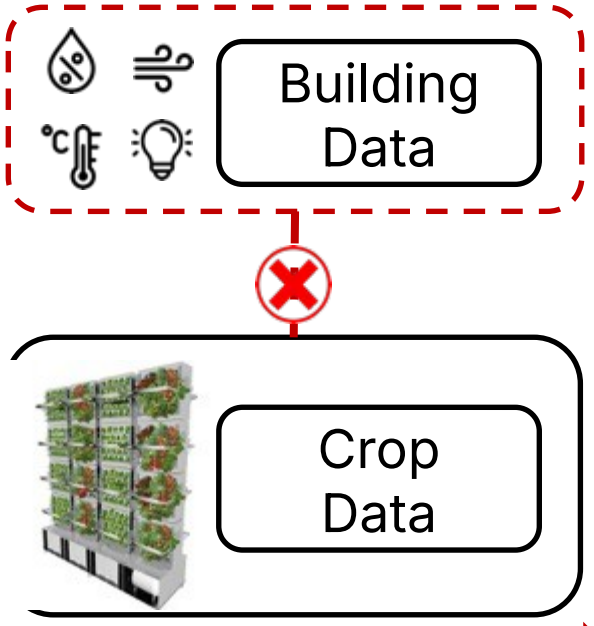
Farm systems do not collect building data



Container farms not built for scale



Building solutions are too general



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Solution: AI Platform for AgTech Facilities





Sense

Optimize



Control



Singapore patent pending: 10202105064T

1. Connected

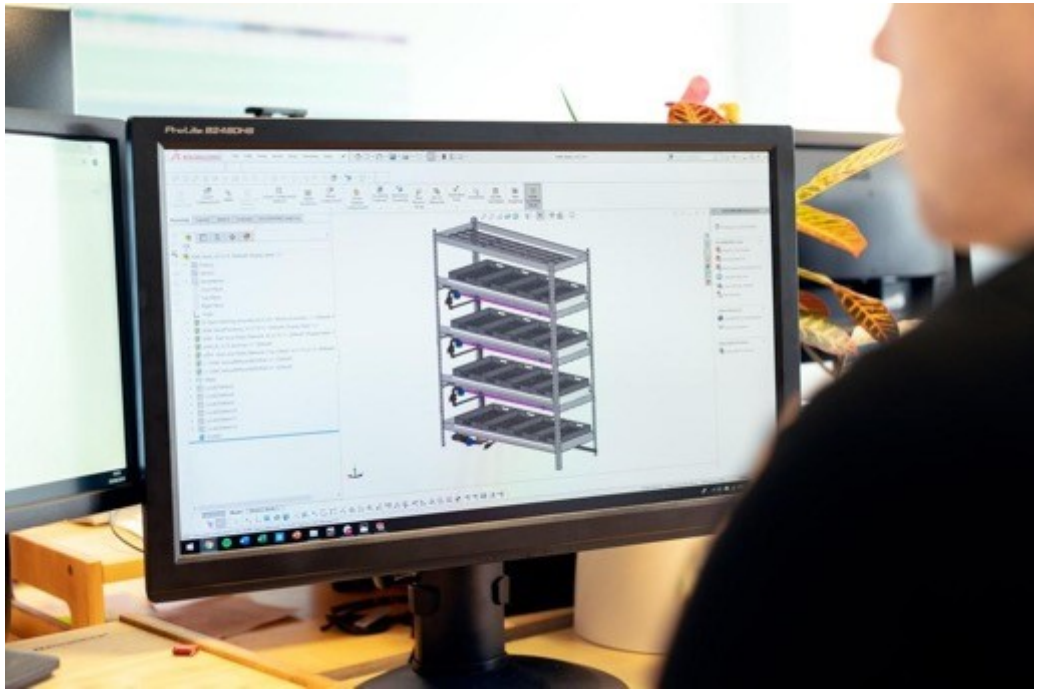
Integrated with essential equipment and control systems

Seamless Building Integration

Connect the indoor farming system to the existing building management system – or have Ambient implement them based on your requirements. Enjoy full connectivity across the farm and perform real-time environmental adjustments based on crop.

2. Centralized

Important data in one easy-to-access place



Monitoring In Real-Time

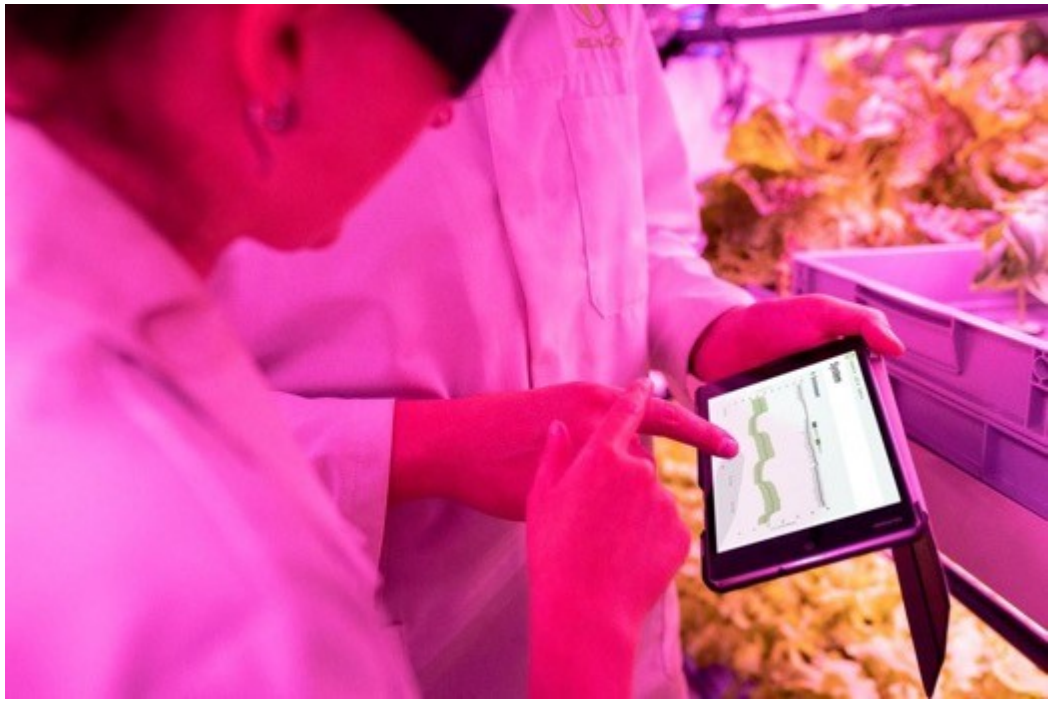
Access real-time data through online dashboards and securely archive all critical operational information. Start measuring most important crop parameters like temperature, relative humidity, and CO2.

3. Automated

Leverage combined power of AI and Human Experts

Grow With Data Insights

Leverage latest machine learning models to improve the harvest and reduce operational costs. Access curated insights by our network of crop-scientists who can help you tun—in your farm operations, enabling better crop health and more accurate harvest timing predictions.



Data-Driven Automation

Climate Data



Crop Data



Equipment Settings

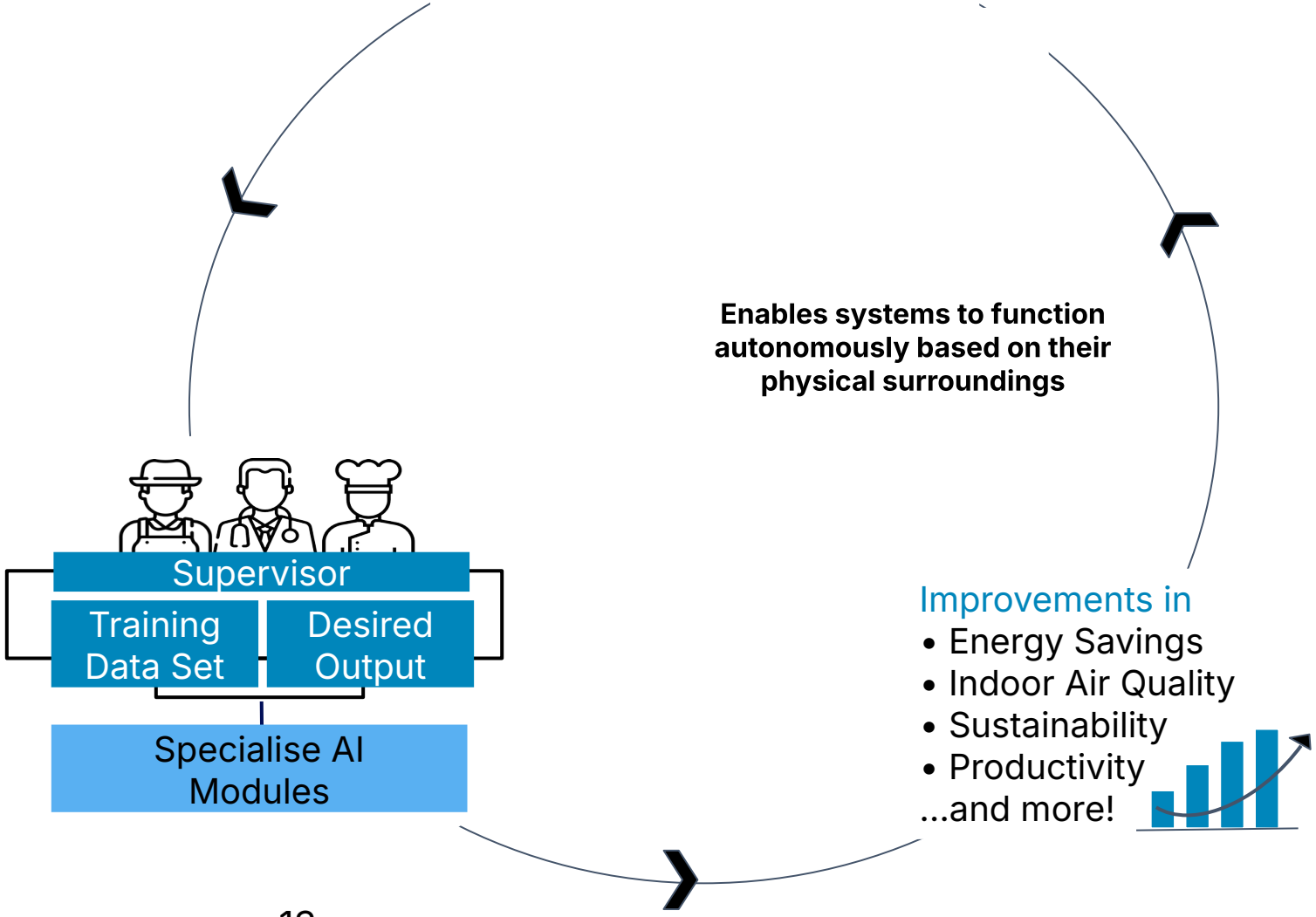


	Parameter
Monitor	Air Temperature
	Relative Humidity
	CO2
	VOCs

	Parameter
Crop Insights	Soil Temperature
	Soil Moisture
	pH level
	Electrical conductivity (EC_w)
	Crop specific parameters

Sample Action
• Adjust temperature • Adjust ventilation • Adjust lights • Turn on fogging machines • Turn on CO2 regulator • Balance irrigation and humidity levels

Facility Management beyond Energy Savings



Private and Confidential



+24%

energy savings, man-hour savings, enhanced crop performance

Break-even in

6 to 12 months

Usage-based Pricing Plan

One-time Implementation Cost
Installation of sensors, networks, and equipment integration

by Project Basis

Basic
SGD $500 per week

- Custom data analytics

2-3 Data Feeds

Advanced
SGD $1,000 per week

- Customer data analytics
- Customized AI module

4-5 Data Feeds

Premium
SGD $2,000 per week

- Customer data analytics
- Industry vetted AI module
- Cybersecurity enhancements

6-7 Data Feeds

Traction



Working with four AgTech Companies in Singapore

Approved by industry leaders in building technology

Our Team



- **MComp, Computer Science**
  National University of Singapore
- **BASc, Civil Engineering, Sustainable Energy**
  University of Toronto
- **Serial Entrepreneur**
  5 years experience in cleantech and deep tech

- **BBM, Finance**
  Singapore Management University
- **Strategy and Fundraising**
  3 years experience across e-commerce, gaming, and payments

- **Specialist Diploma, Communications**
  Nanyang Polytechnic / Temasek Polytechnic
- **Creative**
  3 years experience across graphic design and content creation for various brands

Work With Us






Indoor Farms
interested in cutting-edge automation solutions

Advisors
with expertise in Agri-tech / Artificial Intelligence / Building Technology

Technology Vendors
specializing in sensor manufacturing and system integration

Thank You

Visit us at: www.ambientsystems.ai

